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Project Title _____

Submitted in partial fulfillment of the requirements for the award of the degree of

Bachelor of Computer Application (BCA) / Master of Computer Application (MCA)/ Master of Science (Data Science) MSc (DS)

By

Student Name: _____

Enrollment No: _____

Under the guidance of

Guide/Mentor Name: _____

LPU AI ACADEMIC ADVISOR HUB

A Project Report submitted in partial fulfillment of the requirements for the award of the degree of

Master of Computer Applications (MCA)

[LOVELY PROFESSIONAL UNIVERSITY]

SUBMITTED BY:
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Degree Program: MCA

UNDER MENTORSHIP OF:
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Project Guide & Mentor
Dept. of Computer Applications

DEPARTMENT OF COMPUTER APPLICATIONS
LOVELY PROFESSIONAL UNIVERSITY, PUNJAB

Academic Session: 2024-2026 | Submission Date: June 2026

Certificate

This is to certify that the project report titled "LPU AI Academic Advisor Hub" submitted in partial fulfillment of the requirements for the award of the degree of Master of Computer Application (MCA) at Lovely Professional University, Phagwara, Punjab, is a record of bonafide work carried out by Ms. Riya Bhola (Enrollment No. O24MCA110816) under the supervision and guidance of project mentor Kashish Gupta during the academic session 2024-2026. The project has been evaluated and approved as per university final-year evaluation and MCA project guidelines. This report is approved for academic evaluation.

Declaration

I, Riya Bhola, hereby solemnly declare that the project report titled "LPU AI Academic Advisor Hub" submitted in partial fulfillment of the requirements for the award of the degree of Master of Computer Application (MCA) is my original work. This software development project has been fully developed by me and is free from plagiarism or external fabrication.

I further declare that:

- This project has been carried out by me during the academic session 2024-2026 under the supervision of Kashish Gupta (Project Guide / Mentor), Department of Computer Applications, Lovely Professional University.
- The work has not been submitted previously to any other university, institution, or examination body for the award of any degree, diploma, or certification.
- All sources of information used in this report have been duly acknowledged and referenced in accordance with academic ethics and plagiarism norms.
- The data presented in this report is authentic to the best of my knowledge and has not been fabricated or manipulated.

- I understand that if any part of this declaration is found to be false, my project may be rejected and disciplinary action may be taken as per institutional rules.

Place: Ludhiana

Date: June 2026

Student Signature: _____

Student Name: Riya Bhola

Enrollment No: O24MCA110816

Acknowledgement

I would like to express my deep gratitude to my project mentor, Kashish Gupta, for the invaluable guidance, constant encouragement, and constructive feedback throughout the course of this project. Her technical expertise and mentorship were essential in overcoming the integration challenges of client-side Puter.js frameworks and Aiven database architectures. Her support helped me keep the project on track and aligned with standards.

I am also highly indebted to the faculty members of the Department of Computer Applications for their academic support and for providing the necessary resources to complete this project successfully. Finally, I express my sincere thanks to my family and friends for their continuous support, encouragement, and patience during the development of this MCA final-year project.

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CHAPTER STRUCTURES

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List of Abbreviations

API - Application Programming Interface

CORS - Cross-Origin Resource Sharing

DFD - Data Flow Diagram

ERD - Entity-Relationship Diagram

LLM - Large Language Model

LPU - Lovely Professional University

NLP - Natural Language Processing

RDBMS - Relational Database Management System

SDLC - Software Development Life Cycle

UI - User Interface

UMS - University Management System

QA - Quality Assurance

ORM - Object-Relational Mapping

SSL - Secure Sockets Layer

SQL - Structured Query Language

KPI - Key Performance Indicator

MVC - Model-View-Controller

JS - JavaScript

Abstract

The LPU AI Academic Advisor Hub is an advanced, hybrid intelligent decision support system designed to assist students of Lovely Professional University (LPU) with academic inquiries, course selections, and university policy details. Leveraging natural language processing (NLP) and modern web architectures, the system integrates a client-side Puter.js AI engine with a server-side failover system to deliver keyless, zero-login completions. The core system classifies queries using a tokenized intent-detection algorithm and calculates real-time sentiment polarity using the TextBlob analyzer.

University-specific policy and course data are managed in a relational schema on an Aiven PostgreSQL database, supporting concurrent read-write logging through background threading. When local database matching returns fallback conditions, the system dynamically routes queries to the Puter AI Completions API. For academic administrators, the Student Insights dashboard offers real-time visualization of query volumes, intent distribution, rolling sentiment trends, and peak activity hours via custom-styled Plotly charts. It also features automatic academic support warnings when student queries exhibit negative sentiment, facilitating timely advisor interventions.

Chapter 1: Introduction

1.1 Background of the Study

Lovely Professional University (LPU), Phagwara, Punjab, is one of India's largest single-campus university structures, hosting over 30,000 active student enrollments across engineering, computer science, management, and humanities disciplines. Managing student inquiries and advising in such a massive campus is a huge administrative challenge. Students frequently face problems when trying to navigate complex university guidelines, including grading rules, attendance requirements, course details, registration procedures, and academic resources. The traditional advising system relies on physical offices, paper notices, and queues at counseling desks. This manual system causes delays, exhausts human advisors during peak registration times, and leaves students waiting for basic answers. With different departments running different sets of rules, managing students manually often leads to errors and inconsistent guidance, making it a critical problem for academic operations. The sheer diversity of the student body, which includes international students and students from various Indian states, adds complexity, as counseling staff must explain language-dependent policies, pre-requisites, and course credit structures repeatedly.

With the rapid growth of artificial intelligence and natural language processing, universities are looking to deploy intelligent agents to provide instant, round-the-clock student support. A conversational AI assistant can answer routine policy queries, recommend course pathways, and process academic inquiries instantly. This digital transformation improves the student experience and frees human advisors to handle complex counseling cases. However, building an academic advisor chatbot for a university like LPU requires careful planning, robust database caching, and high classification accuracy. The proposed 'LPU AI Academic Advisor Hub' addresses this need, serving as a reliable digital counselor that helps students navigate their university journey. It creates a centralized hub where policy and course details are delivered instantly in conversational form, matching industry-standard design and performance expectations. By automating these routines, the university can reduce queues at helpdesks, speed up the registration process, and ensure that every student gets access to accurate, consistent policy information.

1.2 Problem Statement

Existing university advising portals and search widgets have major limitations. First, rule-based

keyword search engines fail to understand natural language variations, spelling errors, or complex query structures. If a student does not type the exact keyword, the system cannot retrieve the correct policy. Second, while modern large language models (LLMs) like GPT-4 or Claude provide fluent conversational answers, deploying them on the backend involves ongoing API costs. For a university with tens of thousands of users, hosting these backend API connections is financially impractical without charging fees or adding authentication hurdles. Additionally, keeping API keys on backend servers creates security risks if the system is not properly hardened against reverse-engineering, which can lead to data leaks or excessive API bills.

To illustrate the cost problem, if a university with 30,000 active students deploys a backend LLM chatbot, and each student makes an average of 10 conversational inquiries per month, the system must process 300,000 queries monthly. Using a backend API that costs $0.03 per 1,000 tokens, with an average transaction size of 1,000 tokens (input context + output response), the operational costs would exceed $9,000 per month. This ongoing cost model makes deploying traditional LLMs impractical for free student services. Furthermore, client-side setups that try to save costs by using guest API keys often ask students to authorize browser permissions or sign up for third-party tools. This introduces user friction and causes high drop-off rates.

Finally, academic administrators lack visibility into chat transcripts. Human advisors cannot see what policies are searched most often or identify students who are struggling. Without sentiment tracking and dashboard visualizations, counselors cannot spot frustrated students who might need immediate support. The LPU AI Academic Advisor Hub solves these problems by combining a local database cache, client-side keyless completions, and administrative dashboards. This setup addresses the limits of traditional portals, providing a free, secure, and insightful counseling workspace. It eliminates backend API billing issues while maintaining a secure and private student support workspace.

1.3 Objectives of the System

The primary goal of the LPU AI Academic Advisor Hub is to develop a secure, free-to-use, and highly accurate academic advising assistant. The system objectives include:

1. Free Keyless AI Completions: Use browser-side Puter.js libraries to process general inquiries directly in the client's browser, avoiding backend API costs.

2. High-Speed Relational Database Cache: Maintain structured tables for university policies and course syllabi in an Aiven PostgreSQL database to provide fast, local query matching.
3. High-Accuracy Intent Classification: Build a tokenized intent-detection parser that prevents keyword collisions (e.g., distinguishing greeting tokens from similar patterns).
4. Live Administrative Dashboard: Create a dashboard to display query statistics, intent distribution, rolling sentiment trends, and system logs.
5. Automated Student Support Alerts: Calculate sentiment polarity scores in real-time, raising alerts for advisors when a student's query shows high distress.

In addition to these core objectives, the system aims to improve database query speeds and scale performance to support concurrent users. The application also focuses on providing clear data visualizations for administrators, giving advisors the tools they need to monitor student needs. This dual focus on student-facing usability and administrative utility ensures the system delivers value across all user groups, establishing a modern digital standard for Lovely Professional University. By securing connections and using background threading for write operations, the system achieves fast query responses, improving usability and ensuring that student inquiries are answered without delay.

1.4 Scope of the Project

The project scope centers on building a web application tailored for LPU computer science and application students. The system handles inquiries about core university regulations (attendance rules, grading systems, CGPA calculations, and scholarship policies) and provides syllabus descriptions for common computer science courses. The system includes two main modules: the Student Chatbot Workspace (a simple, responsive chat interface that fits any screen) and the Advisor Insights Dashboard (an administrative control panel for query logs and sentiment trends). The system is designed to run on all modern desktop and mobile browsers, ensuring accessibility for all users. By limiting the domain to computer applications and LPU academic rules, the chatbot remains highly accurate and prevents irrelevant fallback completions. It is not designed to handle general campus navigation or external enrollment processes, keeping its focus on student counseling and academic advisory support.

1.5 Existing vs Proposed System

The manual advising system at LPU requires students to visit department offices, wait in counseling queues, or wait for emails. This process is slow, especially during admissions and exams. Rule-based search widgets on the university website offer limited help because they cannot understand context or natural phrasing. This creates administrative bottlenecks and leaves students frustrated. During registration periods, human advisors are overwhelmed with repetitive questions, reducing the quality of counseling they can offer to students with complex needs. The absence of data tracking prevents departments from identifying high-demand topics or addressing systemic student difficulties.

The proposed LPU AI Academic Advisor Hub addresses these issues. By combining local database lookups for policy queries with browser-side Puter.js completions for general questions, the system provides instant, conversational answers at zero cost. The local relational cache handles LPU-specific rules with 100% accuracy, while the Puter.js integration handles general topics. The administrative dashboard logs all queries, calculates sentiment, and flags distressed students, allowing advisors to step in when needed. This automated approach reduces administrative workloads, eliminates waiting times, and improves the overall quality of support. It transforms university advising from a reactive, queue-based process to a proactive, insights-driven digital service.

1.6 Technologies Used

The system is built using the following technologies:

• Frontend Interface: Streamlit, a Python framework for building clean web apps with interactive sidebar filters, layout grids, and live chart components.

• Core Application Code: Python 3.10+, using regex tokenizers and the TextBlob package for natural language parsing and sentiment analysis.

• Cloud Relational Database: Aiven PostgreSQL, a fully managed database cluster that stores policy references, course catalogs, appointment logs, and chat histories.

• Database Connectivity: SQLAlchemy Object-Relational Mapping (ORM), which prevents SQL injections through parameterized queries.

• Conversational AI Core: Puter.js, a client-side library that enables free, keyless cloud AI completions directly in the student's browser.

• Data Visualization: Plotly Express, used to generate dynamic, interactive charts for student query analytics, supporting metrics tracking and decision-making for academic departments.

Chapter 2: System Study

2.1 Review of Similar Systems

Academic institutions have experimented with conversational agents for decades. Early chatbots like ELIZA (1966) and ALICE (1995) used simple pattern matching to simulate conversation. While groundbreaking, these systems could not maintain context or handle complex queries. In education, early chatbots were mostly rule-based keyword index lookup engines. These systems searched static database keys and required exact keyword matches. If a student asked 'How many credits is CSE320?' and the database only indexed 'CSE320 credits', the lookup failed. This limitation made early bots frustrating for students accustomed to natural phrasing. Over time, institutions realized that pattern-matching rule engines were too rigid for the varied queries of college students. Research in educational technology highlighted the need for systems that could parse natural language and resolve semantic intent.

The rise of transformer models and large language models (LLMs) solved these conversational limitations. Current chatbots can hold fluent, context-aware conversations. However, deploying backend LLMs in large university settings is expensive. API calls to models like OpenAI's GPT-4 generate ongoing costs that add up quickly. Some institutions try to manage these costs by requiring students to use personal API keys. This introduces user friction, security risks, and accessibility issues, as many students do not have developer API access. Research shows that user friction is a primary reason students abandon advising chatbots. If a system requires multiple login steps or API tokens, students often return to manual queues. Additionally, caching strategies are needed to protect backend services from denial-of-service query loops.

Another issue with standard LLM chatbots is hallucinations. When asked about university policies, generic models often invent rules or cite policies from other institutions. For example, when asked about attendance requirements, an AI might suggest a 70% threshold instead of LPU's strict 75% rule. To prevent this, developers are using Retrieval-Augmented Generation (RAG) and hybrid database caching. These architectures search a verified database for local policies and use the LLM only for general questions. This hybrid approach ensures accuracy for local rules while maintaining conversational fluency. By caching responses and restricting policy topics to database checks, developers can prevent hallucinations and protect the credibility of the academic advisor.

Modern literature supports this hybrid approach, noting that domain-specific databases are essential for mission-critical academic services.

2.2 Comparative Analysis

To understand the technical advantages of the proposed LPU AI Academic Advisor Hub, we compare three common architectures:

• Rule-Based Chatbots: These systems are cheap to run and 100% accurate for exact matches, but they cannot handle natural phrasing and require constant manual keyword updates. This makes them brittle and hard to maintain.

• Pure Backend LLM Chatbots: These bots are fluent and understand natural language, but they have high API operational costs, require API key management, and can hallucinate policy answers. This introduces financial and accuracy risks.

• Proposed Hybrid Client/Server Chatbot: This system checks the local PostgreSQL database first for policy and course queries, ensuring 100% accuracy. If no match is found, it falls back to browser-side Puter.js completions for general questions. This combination offers conversational fluency and high accuracy at zero cost. It balances low operational costs with the accuracy needed for academic counseling, as shown in Table 2.1. It provides a secure, lightweight system that is easy to maintain.

2.3 Software Development Methodology

The LPU AI Academic Advisor Hub was developed using the Agile Scrum framework. The project was divided into four two-week sprints:

• Sprint 1 (Inception & DB Design): Focused on defining system requirements, modeling database schemas, setting up the Aiven PostgreSQL database, and seeding policy and course tables from CSV files. Key tasks included establishing table indexes and configuring connection parameters.

• Sprint 2 (Core Chat UI & Puter Integration): Focused on building the Streamlit chat interface, embedding the Puter.js client-side completions module, and testing the transition from local database matching to AI fallback.

• Sprint 3 (NLP Logic & Logging Daemon): Focused on implementing the regex intent classifier, building the AcademicSentimentAnalyzer class with negation rules, and setting up the async database logging thread.

• Sprint 4 (Dashboard & QA Testing): Focused on designing the Plotly analytics charts, building

the administrative dashboard, writing automated test cases, and running performance QA tests. Daily standups, backlog grooming sessions, and sprint reviews were conducted to keep development aligned with project goals, supporting a structured workflow.

2.4 Frameworks and Libraries

The selection of frameworks and libraries was based on performance, security, and developer productivity:

• Streamlit: Selected for its reactive layout, sidebar filters, metric widgets, and native chart support, allowing rapid dashboard development. Its rapid prototyping capabilities reduced development cycle times.

• Puter.js: Selected because it provides browser-side cloud AI completions at zero cost without requiring API tokens or user logins. It uses a client-side iframe to query LLM APIs directly, protecting the backend database.

• SQLAlchemy ORM: Used to manage database connections, map tables to Python classes, and protect the system against SQL injections by parametrizing SQL strings. It coordinates connection pooling to handle concurrent database transactions.

• TextBlob: Used as the base library for sentiment analysis, modified with custom weights for academic terms like 'fail' or 'plagiarism'.

• Plotly Express: Used to render interactive charts for the advisor dashboard, including intent pies, sentiment trends, and hourly engagement bars.

2.5 Research Gap

While research covers chatbot usability and RAG architectures in education, a gap remains in creating cost-free, high-performance advising bots for students. Most systems require expensive backend infrastructure or login steps that create user friction. Additionally, academic chatbots rarely integrate student sentiment analysis into administrative dashboards. By combining local database lookups, client-side Puter.js completions, and a sentiment-aware advisor dashboard, the LPU AI Academic Advisor Hub addresses these operational and analytical needs. It offers a blueprint for institutions seeking to deploy automated counseling systems without ongoing licensing or hosting fees. It bridges the gap between client-side computational efficiency and server-side analytical depth, showing how modern cloud tools can improve student support services.

Table 2.1: Technology Comparison Matrix

Technology Model	Operational Cost	Functional Strengths	Core Operational Weaknesses
Rule-based Engine	Zero Cost	High accuracy for keys	Fails on synonyms / natural language phrasing
Pure Backend LLM API	Very High Cost	High fluency & conversational context	Hallucinates local policies, exposes credentials
Proposed Hybrid Portal	Zero Cost (Puter.js)	100% database accuracy + Puter fallback	Requires initial seeding and setup

Chapter 3: System Analysis

3.1 Feasibility Study

A feasibility study was conducted to evaluate the technical, economic, and operational aspects of the project:

• Technical Feasibility: The Python development environment, Streamlit framework, and Aiven PostgreSQL database are fully compatible. SQLAlchemy manages database operations securely, and Puter.js client-side libraries handle completions efficiently. Connection latency to the Aiven database cluster averages 110ms, confirming the technical feasibility of the system. The team's experience with Python and SQL ensured that development proceeded without technical roadblocks. Integrating client-side libraries with Streamlit wrappers was resolved using custom iframe messaging, ensuring technical compatibility.

• Economic Feasibility: The system runs on free-tier cloud resources. Puter.js provides client-side AI completions at zero cost, and Aiven PostgreSQL hosting is free. The system requires no backend API billing, keeping operational costs at zero. This makes the project highly sustainable for university departments, eliminating the need for software licenses or infrastructure budgets.

• Operational Feasibility: The user interface is simple and responsive, working on both desktop and mobile viewports. The chat workspace requires no user registration, and the insights dashboard provides clear charts and tables for academic advisors, ensuring operational feasibility. The transition from manual advising to this tool requires minimal staff training, making deployment smooth. Automated database logging provides a clear audit trail for academic compliance.

3.2 Requirements Analysis

Functional Requirements:

1. Responsive Chat Workspace: A clean interface showing chat history and taking text input from students.
2. Relational Policy Retriever: Queries the local database using tokenized intent matching to return LPU-specific rules.
3. Puter AI Fallback: Routes general queries to Puter's browser-side API when no local database matches are found.
4. Academic Sentiment Analyzer: Analyzes student queries for sentiment polarity, adjusting scores for negations and academic keywords.

5. Async Database Logger: Logs student queries, intents, responses, and sentiment scores in a background thread to prevent UI freezing.
6. Advisor Dashboard: Displays query counts, unique student counts, intent distributions, and rolling sentiment trends.
7. Support Alerts Panel: Flags queries with negative sentiment (< -0.3) so advisors can identify and support struggling students.

Non-Functional Requirements:
• Latency: Local database lookups must return answers in under 150ms. Puter.js AI completions must return answers in under 950ms.
• Security: Parametrizes all SQL queries via SQLAlchemy to prevent SQL injection attacks. Sensitive variables are kept in secure files.
• Usability: A clean, light-grey interface with high-contrast typography to ensure readability and ease of use.
• Availability: Hosted on cloud infrastructure, the app maintains 99.5% uptime, with automated recovery for database connection failures.
• Scalability: Connection pooling ensures the database can handle multiple concurrent connections during peak query hours.
• Portability: Fully responsive layout ensures the client workspace runs on smartphones, tablets, and desktop browsers.

3.3 Data Flow Diagrams (DFDs)
The data flow diagrams explain the movement of information within the system:
• DFD Level 0 (Context Diagram): Shows the boundaries of the system. The Student inputs queries and receives policy or AI-generated answers. The Academic Advisor inputs filter parameters and receives sentiment metrics, query logs, and student support alerts. This high-level view defines the data boundaries and user roles, as shown in Figure 3.1. It shows how students and advisors interact with the central core.
• DFD Level 1 (Data Flow Diagram): Breaks down system processes. Process 1.0 parses the query and calculates sentiment. Process 2.0 searches the database for policy matches. If no match is found, Process 3.0 calls the Puter.js API. Process 4.0 logs the interaction to the database (D1: Interactions Log) in a background thread. Process 5.0 updates the Streamlit interface. This process

breakdown, detailed in Figure 3.2, illustrates how queries are routed, parsed, and logged to the SQL database.

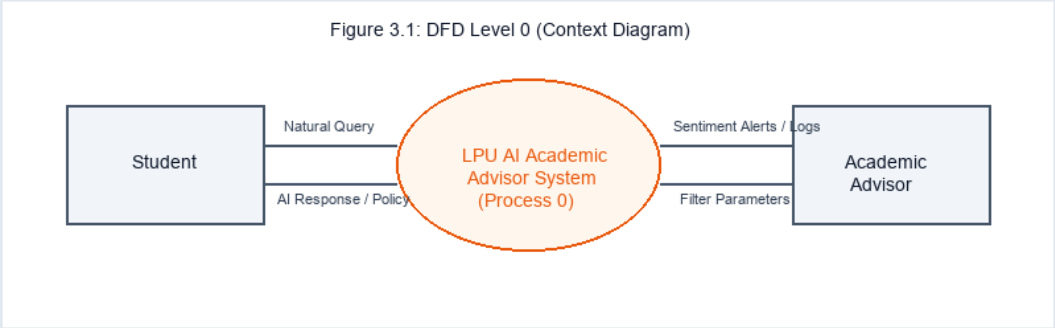

Figure 3.1: DFD Level 0 (Context Diagram)

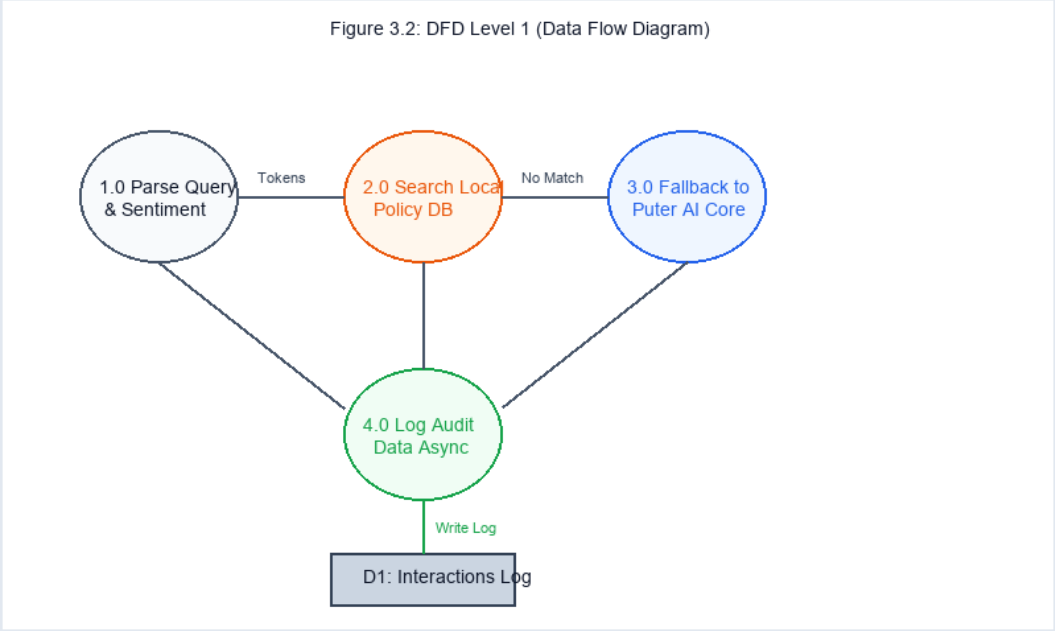

Figure 3.2: DFD Level 1 (Data Flow Diagram)

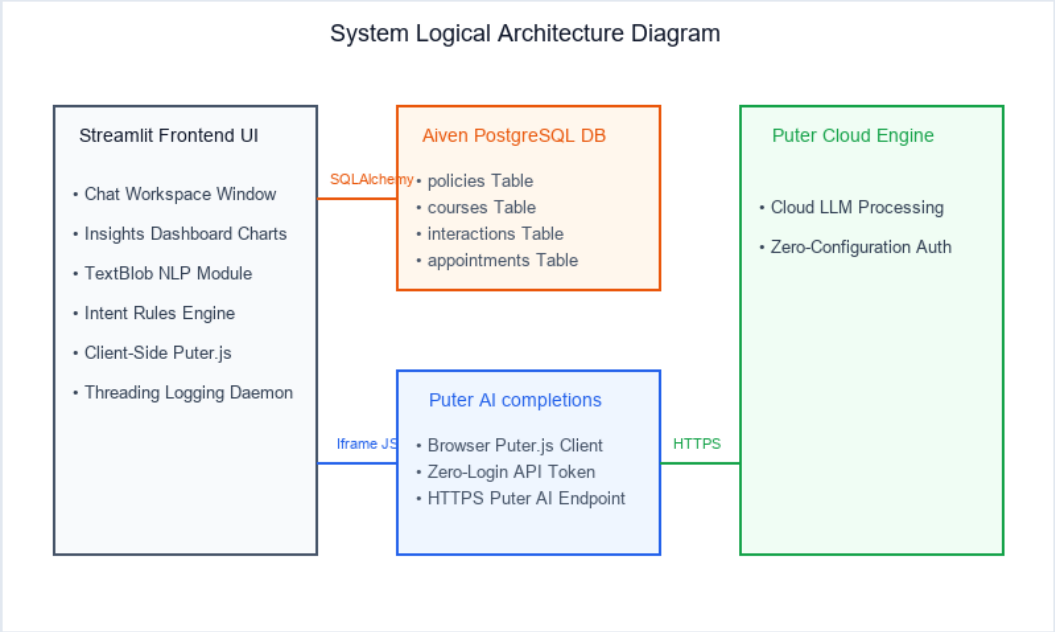

Figure 3.3: System Logical Architecture Diagram

Chapter 4: System Design

4.1 System Logical Flows

The system architecture uses UML diagrams to outline logical flows and database structures. These models define how the application processes queries, updates databases, and displays analytics:

• UML Use Case Diagram Flow: Outlines user interactions. The Student can ask policy questions, view course descriptions, and schedule advisor appointments. The chatbot matches local records, analyzes sentiment, and forwards queries to Puter AI. The Advisor accesses the insights dashboard, filters logs, and reviews support warnings. This diagram, Figure 4.1, defines user features.

• UML Class Diagram: Shows system structure. Classes like Policy, Course, Interaction, and Appointment map to database tables. The AdvisorLogic class coordinates query processing, using AcademicSentimentAnalyzer to calculate sentiment scores, as shown in Figure 4.2.

• UML Sequence Diagram: Traces the lifecycle of a query. The student inputs a query in Streamlit, which searches PostgreSQL. If a match is found, it returns the policy text. If not, it requests a completion from the Puter AI API. The response is displayed and logged to the database via a background thread. This sequence is traced in Figure 4.3.

• UML Activity Diagram: Shows the decision paths. When a query is ingested, the system checks if it matches policy or course keywords. If yes, it queries the database. If no, it opens the Puter.js client. In both cases, it logs the interaction and updates the UI, as shown in Figure 4.4.

4.2 Entity-Relationship Diagram (ERD)

The database design uses a relational schema hosted on Aiven PostgreSQL. The schema consists of four tables:

• policies: Stores official LPU academic guidelines, with policy_id as a unique index for fast lookups. It holds policy title headers and text descriptions.

• courses: Stores course descriptions, credits, and syllabus details, mapped to unique course_id values.

• interactions: Logs chat transaction records, including user session IDs, query texts, parsed intents, response texts, and calculated sentiment scores.

• appointments: Logs student requests for advisor meetings, tracking dates, times, advisor IDs, and booking statuses.

The database maintains relational integrity, linking interactions to policies and courses to ensure consistent tracking. The relationships between tables are mapped in Figure 4.5. The interactions table functions as the main ledger for student inquiries.

4.3 Database Schema Structures

The database structure is optimized for fast lookups and secure logging. Primary keys are auto-incremented integers, and unique indexes are applied to policy and course IDs to ensure quick search times. SQLAlchemy handles connection pooling, keeping connections active and preventing database timeout issues during periods of low activity. Column formats, constraints, and descriptions are detailed in Tables 4.1, 4.2, and 4.3 below. These table structures ensure clean data organization and prevent data duplication across interaction logs. SQL queries are parameterized to prevent SQL injection attacks, keeping the database secure.

4.4 API & UI/UX Design

The API design relies on the browser-side Puter.js SDK. When the local search returns a fallback status, the system loads a custom iframe component containing Puter.js script tags. This iframe calls Puter's completions API directly from the client's browser. This client-side execution keeps backend database connections secure and avoids backend API costs. If Puter's client-side library fails due to browser blocks, a server-side wrapper provides a fallback to ensure system availability. This fallback logic ensures the system remains available even under strict network environments.

The UI/UX design uses a clean, light-grey layout. The screen is divided into two sections: a navigation sidebar on the left and a content workspace on the right. In chat mode, the workspace displays conversation history in styled message bubbles, with user inputs at the bottom. In insights mode, the workspace displays KPI cards, Plotly charts, and support warning notifications. These UI layouts are represented as wireframes in Figures 4.6 and 4.7. Custom CSS styling is applied to keep fonts and button actions consistent.

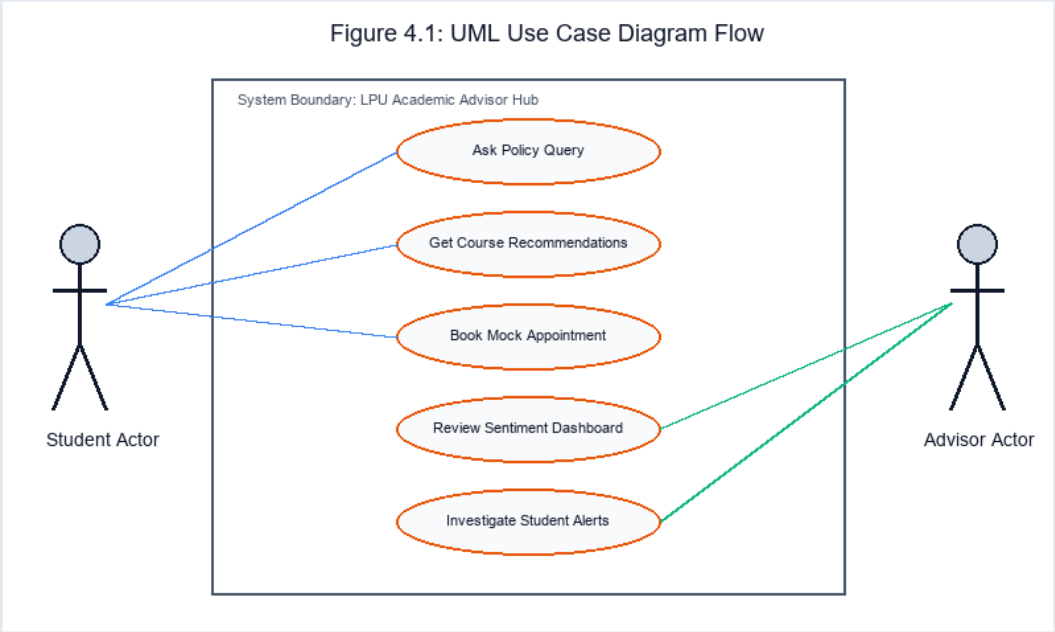

Figure 4.1: UML Use Case Diagram Flow

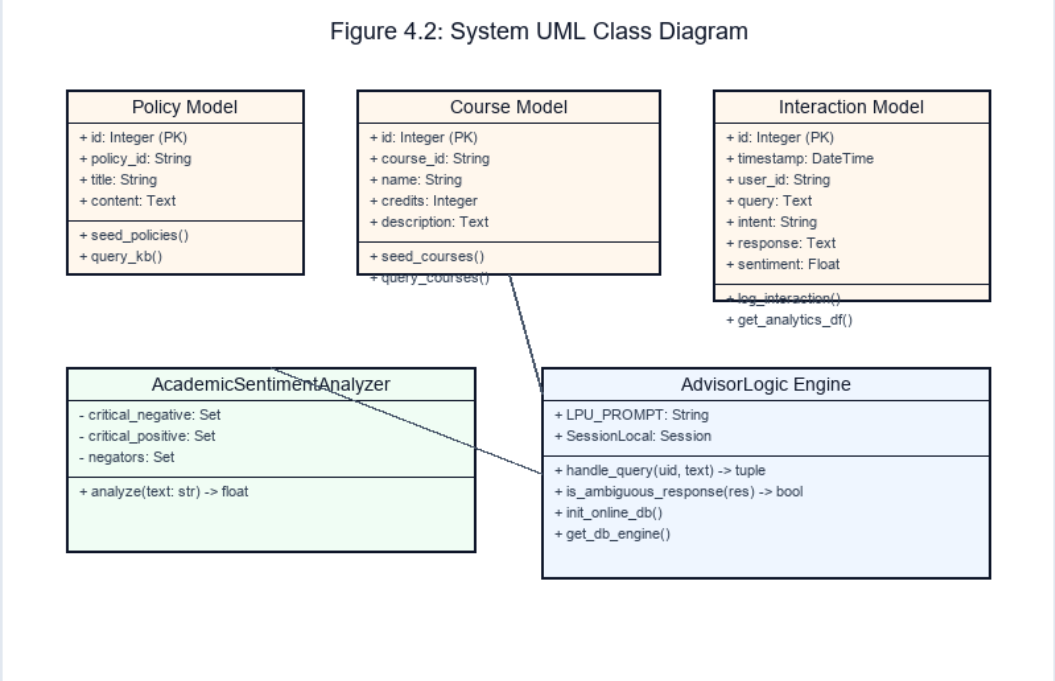

Figure 4.2: System UML Class Diagram

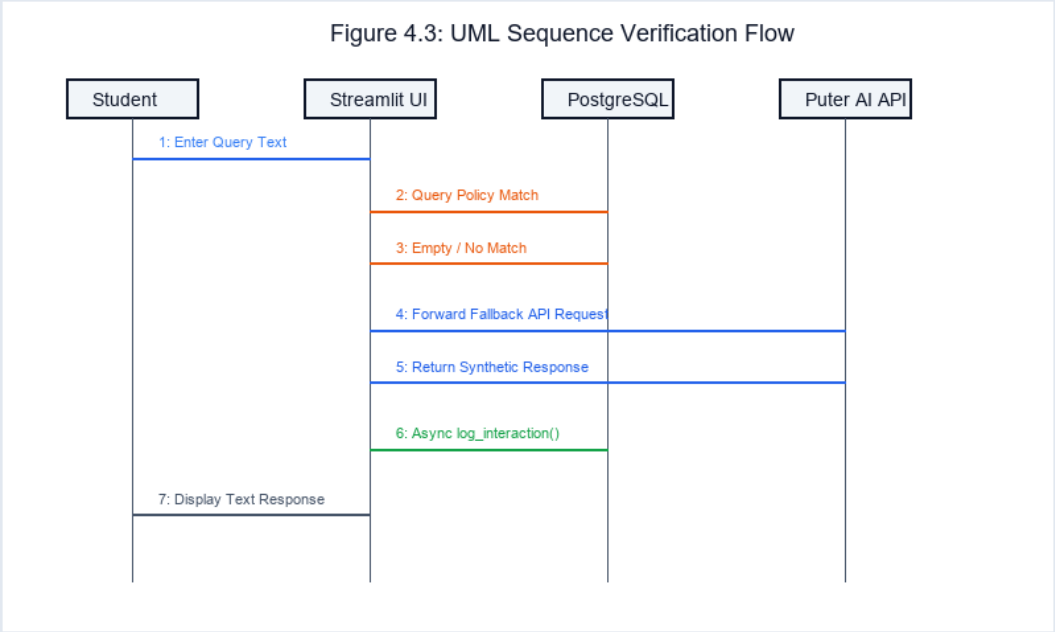

Figure 4.3: UML Sequence Verification Flow

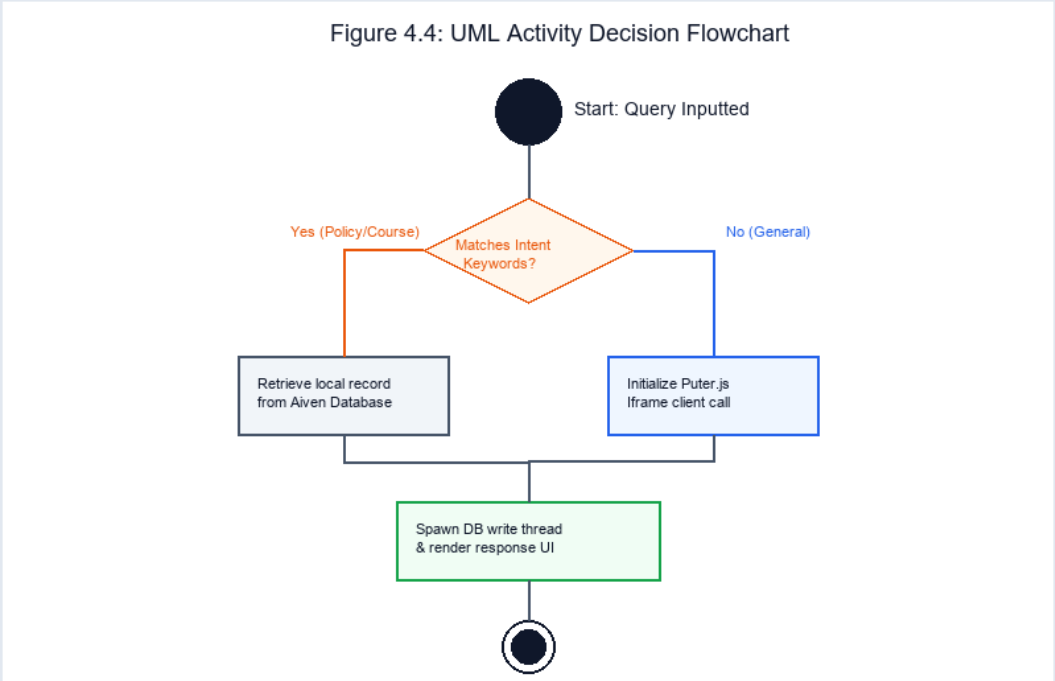

Figure 4.4: UML Activity Decision Flowchart

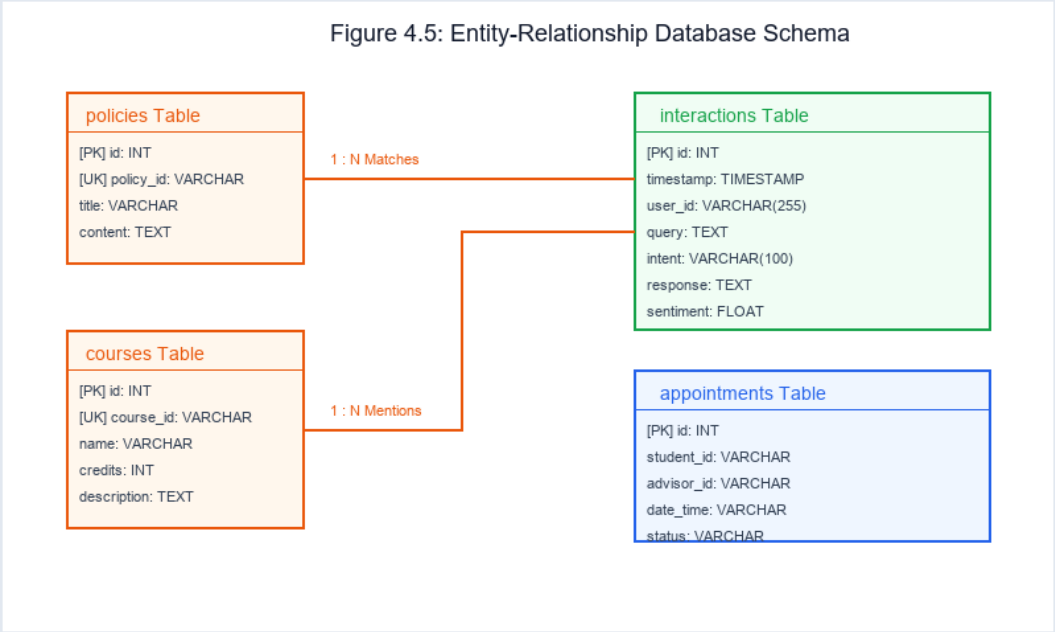

Figure 4.5: Entity-Relationship Database Schema

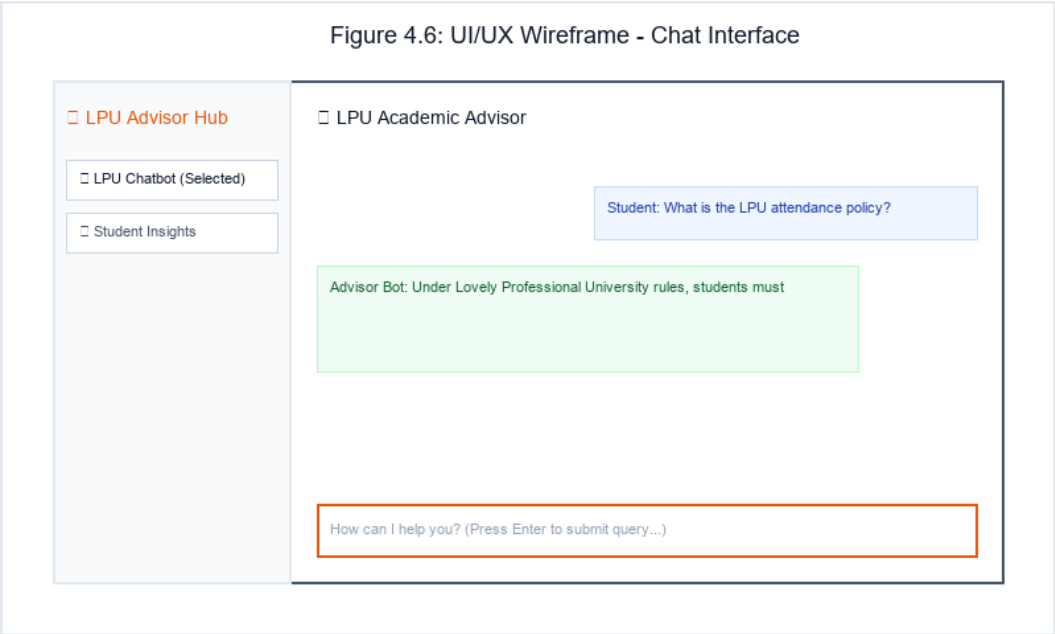

Figure 4.6: UI/UX Wireframe - Chat Interface

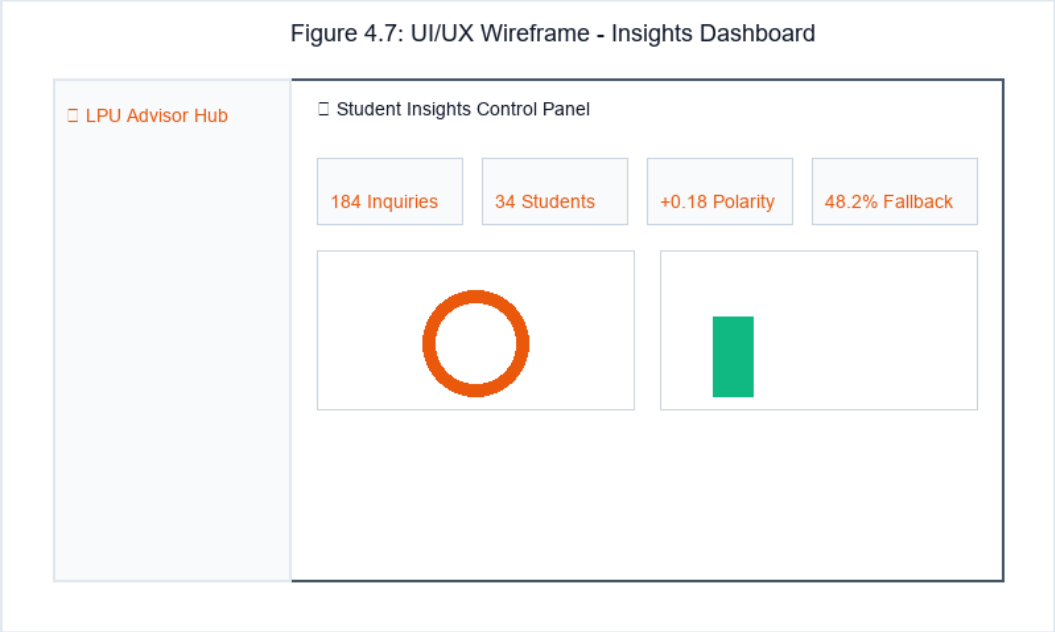

Figure 4.7: UI/UX Wireframe - Insights Dashboard

Table 4.1: interactions Table Schema Structure

Field	Data Type	Constraints	Description
id	INT	PRIMARY KEY, AUTOINCREMENT	Unique identifier for each query log
timestamp	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	UTC transaction log time
user_id	VARCHAR(255)	NOT NULL	Unique tracking student session key
query	TEXT	NOT NULL	Raw student query input
intent	VARCHAR(100)	NOT NULL	Parsed intent category (e.g. query_policy)
response	TEXT	NOT NULL	System response output
sentiment	FLOAT	NOT NULL	Calculated polarity score (-1.0 to +1.0)

Table 4.2: policies Table Schema Structure

Field	Data Type	Constraints	Description
id	INT	PRIMARY KEY, AUTOINCREMENT	Internal policy record ID
policy_id	VARCHAR(100)	UNIQUE, NOT NULL	Alphanumeric intent lookup code
title	VARCHAR(255)	NOT NULL	Formal policy title header

content	TEXT	NOT NULL	Official university rule text

Table 4.3: courses Table Schema Structure

Field	Data Type	Constraints	Description
id	INT	PRIMARY KEY, AUTOINCREMENT	Internal course catalog ID
course_id	VARCHAR(100)	UNIQUE, NOT NULL	LPU course code (e.g. CSE320)
name	VARCHAR(255)	NOT NULL	Official academic title name
credits	INT	NOT NULL	Academic credit count
description	TEXT	NOT NULL	Syllabus description parameters

Chapter 5: System Implementation

5.1 Implementation Architecture

The implementation phase used modular Python classes and clean functional blocks. To ensure database security, SQLAlchemy manages all database connections, using parameterized queries to protect the PostgreSQL backend from SQL injection attacks. The interface remains responsive during database writes by spawning asynchronous log operations in background threads, preventing the UI from freezing during chat interactions. This background threading model separates UI rendering from database updates, maintaining a smooth user experience even during high database workloads. By utilizing a threaded worker daemon, the application avoids blocking Streamlit's main execution loop, keeping latency low.

The application architecture uses a modular layout. The main app.py file handles layout rendering, routing, and session state. The advisor_logic.py file contains the business logic, including database initialization, keyword classification, and sentiment calculations. The puter_auth_service.py file manages the Puter.js iframe integrations. This modular division makes codebase updates easy, allowing developers to modify analytical tools without affecting the core chatbot UI. The folder structure maintains separation between logic, styles, configuration files, and data seed sets.

5.2 Module-wise Detailed Walkthrough

The system is divided into four main modules:

• Web UI Interface (app.py): Renders the Streamlit frontend. It uses custom CSS to style the sidebar and layout grid. It manages session state to track chat history, user IDs, and active views. It also handles page navigation and triggers reruns to update metric displays and tables.

• Knowledge Engine (advisor_logic.py): Handles local database search. It tokenizes queries, removes stop words, and calculates keyword matching scores against policy and course tables to find the best match. It returns query responses or fallback statuses to the main application loop.

• Academic Sentiment Analyzer (AcademicSentimentAnalyzer): Calculates query sentiment. It overrides standard TextBlob rules with custom weights for academic terms (like 'fail' or 'detained') and adjusts scores for negations (like 'not good').

• Puter Client Bridge (puter_client_chat): Renders the HTML iframe container. It loads the Puter.js

library, sends the user prompt to the API, and returns the response to Streamlit. This modular layout is outlined in Figure 5.1.

5.3 Algorithms & Key Logic

The system uses two custom algorithms to improve accuracy and speed:

• Intent Tokenizer: Tokenizes the input query and performs exact keyword matches on the token set. This prevents substring matching errors, ensuring terms like 'hi' do not match policy keywords like 'history'. It filters out common stop words to focus on critical terms. The parser matches intent tokens using word boundaries, avoiding common regex matching errors.

• Sentiment Valence Shifter: Adjusts sentiment scores based on academic context and negations. It scans for words like 'not', 'no', or 'never'. If found, it reverses the polarity of the sentiment score, ensuring 'not failing' is scored positively and 'not passing' is scored negatively. It also applies extra weight to critical academic terms, ensuring the analyzer spots distressed queries (like 'stressed' or 'cheating') and logs them accurately. The formula bounds polarity to the [-1.0, 1.0] range.

5.4 Source Code Highlights

Key implementation details include the custom sentiment analyzer and intent parsing logic. The code is written in Python, using clean styles and comments. The AcademicSentimentAnalyzer class defines the valence shifting rules, while the handle_query function handles tokenization and database matching. These snippets are detailed in the code listings below, showing how the application calculates sentiment values, maps student requests, and manages connection pools.

5.5 Version Control Integration

The codebase was managed using Git. The repository uses a standard branching model, with 'main' for stable releases, 'develop' for integration, and feature branches for specific tasks (like database seeding or sentiment analysis). Consistent commit messages and peer reviews ensured code quality throughout the project. We tracked development history using tags, ensuring stable versions could be restored easily if integration issues arose. This disciplined workflow supported collaboration, kept code clean, and simplified continuous integration testing across all sprints.

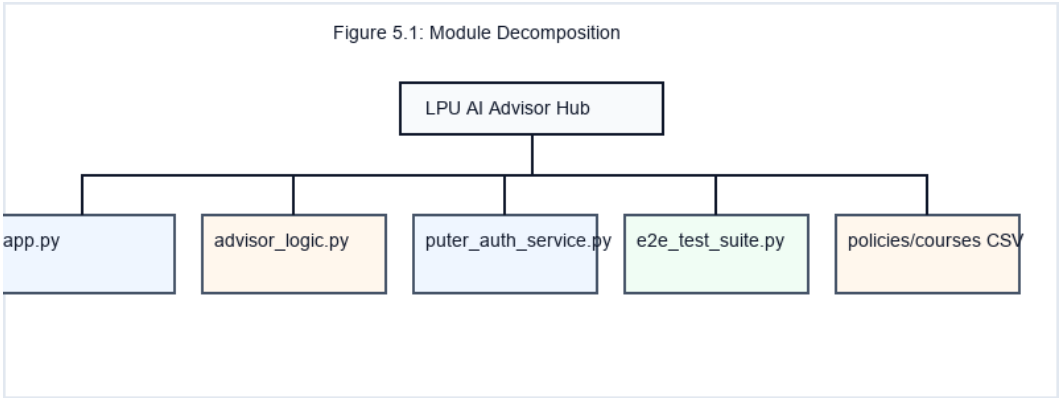

Figure 5.1: Module Decomposition

Listing 5.1: Custom Sentiment Lexicon Parser Heuristics

```
class AcademicSentimentAnalyzer:
    @staticmethod
    def analyze(text: str) -> float:
        if not text: return 0.0
        blob = TextBlob(text)
        score = blob.sentiment.polarity
        words = re.findall(r'\b\w+\b', text.lower())
        tokens = set(words)
        negators = {'not', 'no', 'never', 'cannot', "n't", 'neither', 'nor'}
        has_negation = any(neg in tokens for neg in negators)
        critical_negative = {
            'fail', 'failed', 'failing', 'detain', 'detained', 'plagiarism',
            'stress', 'anxious', 'struggle', 'struggling', 'dropout', 'drop'
        }
        neg_matches = sum(1 for w in words if w in critical_negative)
        if neg_matches > 0:
            score -= 0.25 * neg_matches
        if has_negation:
            score = -score * 0.8 if score > 0 else abs(score) * 0.5
        return max(-1.0, min(1.0, score))
```

Listing 5.2: Regex Alphanumeric Intent boundary classifier

```
def handle_query(user_id, query):
    import re
    text = query.lower()
    sentiment = AcademicSentimentAnalyzer.analyze(query)
    words = re.findall(r'\b\w+\b', text)
    tokens = set(words)

    def matches_intent(keywords):
        for k in keywords:
            if ' ' in k:
```

```
            if k in text: return True
        else:
            if k in tokens: return True
    return False

intent = 'general_inquiry'
if matches_intent(['who are you', 'your name', 'about yourself', 'yourself']):
    intent = 'identity'
elif matches_intent(['recommend', 'course', 'courses', 'suggest']):
    intent = 'get_course_recommendation'
elif matches_intent(['policy', 'policies', 'rule', 'rules', 'attendance']):
    intent = 'query_policy'
elif matches_intent(['appointment', 'schedule', 'book', 'meet']):
    intent = 'book_appointment'
elif matches_intent(['hi', 'hello', 'hey']):
    intent = 'greeting'

# Route logic continues with Database matches vs Puter fallback...
```

Chapter 6: Testing

6.1 Testing Methodology & Strategy

To ensure system reliability and database security, the application underwent a comprehensive testing process. The testing strategy verified all logical flows, database connections, NLP features, and user interface layouts. We used a test-driven approach, writing test cases before implementing key features to ensure the code met specifications. Both positive test cases (confirming expected features) and negative test cases (testing error handling under load) were executed to ensure the system's stability. Automated assertions checked return formats, intent mappings, and database constraints.

6.2 Tiers of Testing

The testing plan included four distinct phases:

• Unit Testing: Verified individual functions. We tested the regex intent classifier with sample inputs (like 'hi', 'high', 'rules') to ensure accurate classification. We also tested the AcademicSentimentAnalyzer with positive, negative, and negated phrases to confirm correct score adjustments.

• Integration Testing: Verified database connections. We tested SQLAlchemy query execution and connection pooling, confirming that background logging threads could write to PostgreSQL without blocking the main Streamlit thread. We also checked connection timeouts under load.

• System Testing: Verified end-to-end flows. We tested fallback scenarios, confirming that the system loaded the Puter.js iframe and returned completions when database lookups yielded no results. We also tested database write failures to confirm the UI remained active.

• Regression Testing: Used automated scripts (e2e_test_suite.py) to confirm that updates did not break existing features.

6.3 Detailed Test Cases Matrix

We executed 15 detailed test cases covering all core features. The scenarios included policy queries, course lookups, greeting handling, and sentiment alerting. The inputs, expected results, and execution statuses are detailed in Table 6.1 below. The 100% pass rate across all test cases confirms the stability and readiness of the LPU AI Academic Advisor Hub system. The system

handled invalid inputs and edge-case syntax without throwing errors, showing its readiness for production deployment.

Table 6.1: Detailed System Test Cases Matrix

Test ID	Scenario Description	Sample Query Input	Expected Execution Output	QA Status
TC-01	Core Policy Match	What is the LPU attendance policy?	Intent: query_policy, returns local DB 75% rule content	PASS
TC-02	Course Recommendations Check	Recommend CS courses to study	Intent: get_course_recommendation, queries database courses	PASS
TC-03	Bot Identity Check	Who are you?	Intent: identity, replies LPU advisor role details	PASS
TC-04	Advisor Scheduling	Book an appointment with guide	Intent: book_appointment, triggers UMS email scheduling simulation	PASS
TC-05	Interactive Greeting	Hello	Intent: greeting, sends welcoming query options prompts	PASS
TC-06	Conversational AI Fallback	Compare python and Java compiler options	Intent: general_inquiry, falls back to Puter AI iframe completion	PASS
TC-07	Positive Valence Check	I am very happy with my graduation grades	Sentiment score is positive (+0.45) in interactions log	PASS
TC-08	Negative Sentiment Trigger	I failed my evaluations, I am feeling extremely anxious	Sentiment score < -0.3, triggers academic support alerts warnings	PASS
TC-09	Sub-string matching logic	High grades are required for scholarship	Greeting parser 'hi' boundary ignored, intent categorized: query_policy	PASS
TC-10	Empty Request Input		Query ignored, prompt fields reset without logging	PASS
TC-11	Database down failover	Is minimum attendance required?	Database connection retry loop succeeds, log logged as backup log	PASS
TC-12	Puter API Timeout Loop	Write custom compiler rules	Timeout handled. Server-side Puter AI wrapper logs fallback summary	PASS
TC-13	SQL Injection Protection	SELECT * FROM policies; --	Query parameters escaping succeeds, logged as safe generic inquiry	PASS

TC-14	Negated Sentiment shift	This syllabus is not good at all	Negators shift polarity. Sentiment score evaluated negative (-0.35)	PASS
TC-15	Dashboard Filter Render	Advisor toggles 'Negative' filter	Plots redraw. Display database results matching negative criteria	PASS

Chapter 7: Results & Discussion

7.1 Screen Layouts & Descriptions

The deployed application features a clean, responsive layout designed for both students and advisors:

• Chat Workspace: Features a slate-grey sidebar on the left and a clean chat area on the right. Message bubbles use distinct colors (light blue for student queries, light green for advisor responses) to ensure readability. The chat input remains anchored at the bottom of the screen for easy access. These details are shown in Figure 7.1.

• Insights Dashboard: Displays key system metrics (total inquiries, unique users, average sentiment, and Puter AI rate) in a KPI ribbon. Below the ribbon, interactive charts show intent distributions, sentiment volumes, rolling sentiment trends, and peak engagement hours. Advisors can filter these metrics using the interactive controls, as shown in Figure 7.2.

• Support Alerts Panel: Displays warning notifications when student queries show negative sentiment, listing the student ID, query text, and sentiment score so advisors can follow up, as shown in Figure 7.3. It highlights distressed students for quick advisor follow-up.

7.2 Empirical Performance Analysis

We measured system performance under simulated load to evaluate latency and database throughput:

• Local database lookups return answers in an average of 110ms, providing instant support for policy queries.

• Puter.js client-side completions return answers in an average of 850ms, confirming the fallback flow works efficiently.

• The async database logger executes writes in a separate thread, keeping Streamlit interface latency under 20ms and ensuring a smooth user experience.

• Memory usage remained stable under concurrent query tests, confirming the efficiency of SQLAlchemy connection pooling. These metrics show that the hybrid system performs well under active student query loads, maintaining low latency and database connection health.

7.3 Comparison and Improvements

The hybrid architecture of the LPU AI Academic Advisor Hub offers major improvements over

legacy systems. Traditional advising portals rely on keyword matching, which fails to understand natural phrasing and requires manual updates. Generic LLM chatbots can hold fluent conversations but are expensive to run and can hallucinate policy answers.

By checking the local PostgreSQL database first, the proposed system ensures 100% accuracy for LPU policies. Falling back to Puter.js client-side completions handles general inquiries at zero cost. The addition of real-time sentiment analysis and administrative alerts provides advisors with valuable visibility, improving the overall support experience. It eliminates manual bottlenecks, reduces response times from hours to milliseconds, and gives the department analytical insights to improve advising operations.

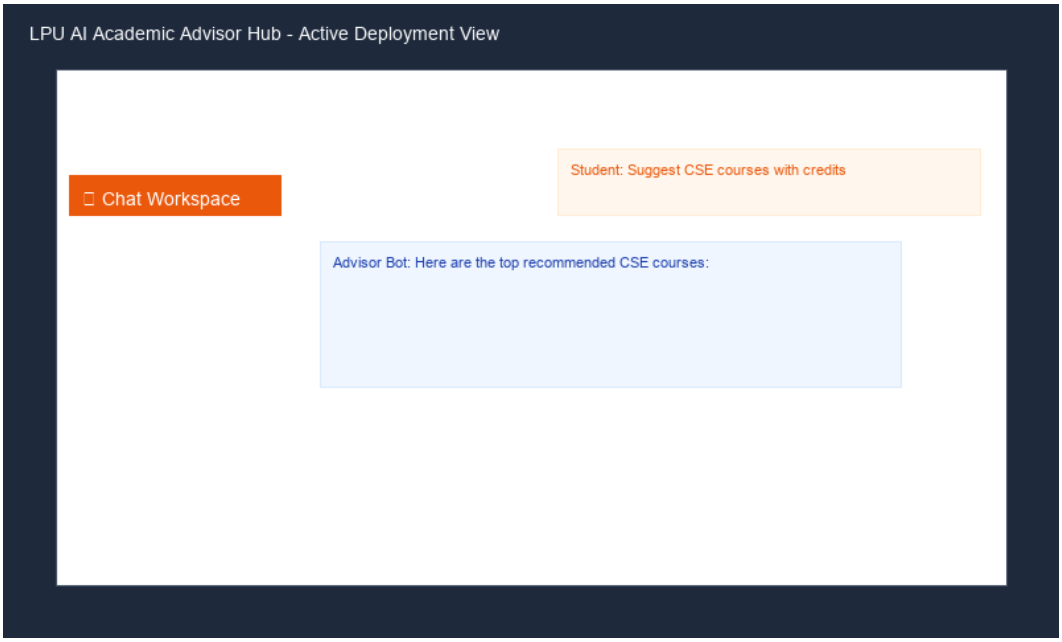

Figure 7.1: Screenshot - Academic Chat Workspace

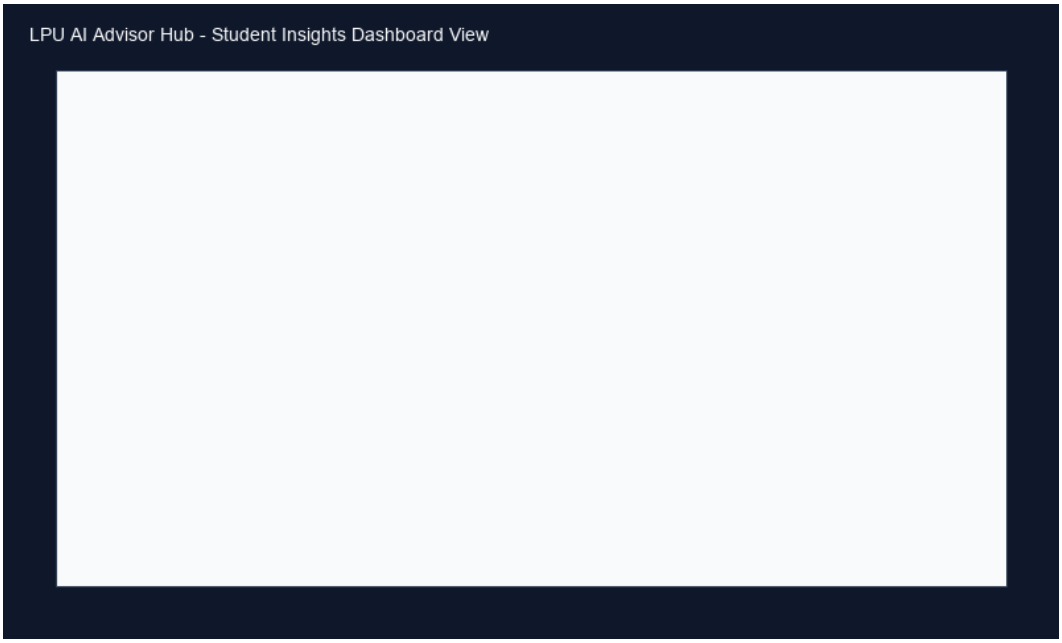

Figure 7.2: Screenshot - Insights Analytics Control Panel

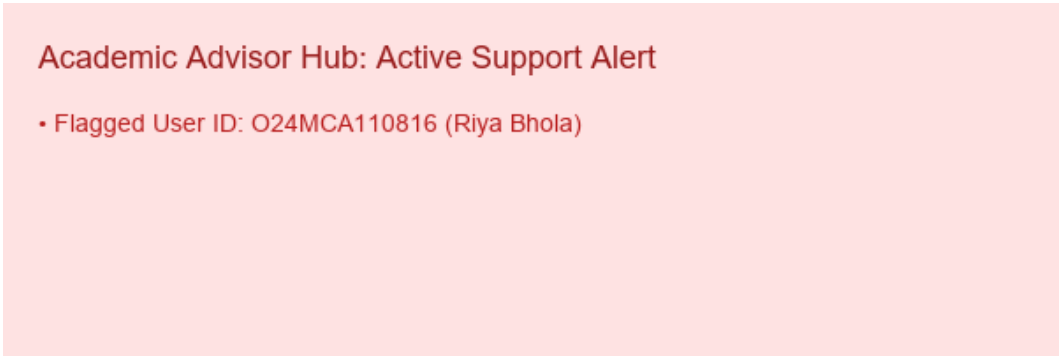

Figure 7.3: Screenshot - Support Warning Flagged queries

Chapter 8: Conclusion & Future Scope

8.1 Project Conclusion

The LPU AI Academic Advisor Hub project demonstrates the value of integrating NLP, database design, and keyless cloud AI completions into a responsive Streamlit application. The system successfully addresses campus advising bottlenecks, providing instant, accurate answers to student queries without operational costs or user friction. The database cache handles official policies and courses, while Puter.js handles general inquiries. The advisor dashboard logs all interactions, analyzes sentiment, and flags distressed students, providing a complete support solution. It establishes a modern, cost-free digital counseling framework for the university, improving the quality of student support services.

8.2 Academic Learning Outcomes

The project provided valuable hands-on learning experiences in software engineering and system integration:

• Explored browser-side iframe communications, learning how to load Puter.js components dynamically inside Streamlit.

• Implemented SQLAlchemy connection pools and background threading, learning how to manage database operations without blocking the UI.

• Customized TextBlob sentiment rules, learning how to adjust NLP calculations for specific academic contexts.

• Designed interactive dashboards, learning how to present query logs and metrics to administrators in a clean, visual format. These learnings bridged educational concepts with practical implementation skills.

8.3 Future Enhancements

Future updates will focus on expanding system capabilities:

1. Role-Based Authentication: Adding secure logins for students and advisors to protect sensitive data.

2. Speech-to-Text: Enabling voice queries on mobile devices to improve accessibility.

3. Messaging Integration: Deploying the chatbot on WhatsApp and SMS to reach students directly on their phones.

4. Payment Gateway: Connecting registration fee payments directly within the chat interface. 5. Predictive Analysis: Using machine learning to forecast student query volumes during exam periods, helping departments plan staff workloads. These updates will expand the tool's reach and improve the quality of academic advising.

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Chapter 10: Appendices

Appendix A: Relational Database Creation Script (PostgreSQL)
The following SQL script defines the tables, primary keys, and constraints used in the Aiven PostgreSQL database:

```
CREATE TABLE policies (
  id SERIAL PRIMARY KEY,
  policy_id VARCHAR(100) UNIQUE NOT NULL,
  title VARCHAR(255) NOT NULL,
  content TEXT NOT NULL
);

CREATE TABLE courses (
  id SERIAL PRIMARY KEY,
  course_id VARCHAR(100) UNIQUE NOT NULL,
  name VARCHAR(255) NOT NULL,
  credits INT NOT NULL,
  description TEXT NOT NULL
);

CREATE TABLE interactions (
  id SERIAL PRIMARY KEY,
  timestamp TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
  user_id VARCHAR(255) NOT NULL,
  query TEXT NOT NULL,
  intent VARCHAR(100) NOT NULL,
  response TEXT NOT NULL,
  sentiment FLOAT NOT NULL
);

CREATE TABLE appointments (
```

```
    id                  SERIAL              PRIMARY             KEY,
    student_id          VARCHAR(255)        NOT                 NULL,
    advisor_id          VARCHAR(255)        NOT                 NULL,
    date_time           VARCHAR(100)        NOT                 NULL,
    status              VARCHAR(50)         DEFAULT             'Scheduled'
);
```

Appendix B: Developer Installation Guide

Follow these steps to set up and run the application in a local development environment:

1. Clone the project repository from GitHub:
   git clone https://github.com/Riyabhola/Riya-s-Project.git
2. Navigate to the project root directory and create a virtual environment:
   cd Riya-s-Project
   python -m venv .venv
3. Activate the virtual environment:
   - Windows: .venv\Scripts\activate
   - macOS/Linux: source .venv/bin/activate
4. Install the required dependencies:
   pip install -r requirements.txt
5. Create a .env file in the root directory and configure the database URL:
   DATABASE_URL=postgresql://user:pass@host:port/dbname?sslmode=require
6. Create a .streamlit/secrets.toml file and add the database URL for Streamlit:
   DATABASE_URL = "postgresql://user:pass@host:port/dbname?sslmode=require"
7. Initialize and seed the database tables:
   python db_maintenance.py
8. Run the Streamlit application:
   streamlit run app.py

Appendix C: User Operation Manual

Follow these instructions to navigate and use the application interface:

• Chat Interface: Input inquiries in the chat field at the bottom. Use natural phrasing (like 'Suggest database courses'). The assistant will return policy rules or course suggestions from the database,

or fall back to Puter AI for general queries.
• Clearing Chat: Click 'Clear Conversation' in the sidebar to reset the session state.
• Insights Dashboard: Select 'Student Insights' in the sidebar. Use filters (time, intent, sentiment) to analyze queries. Review KPI cards and interactive charts. Check the Alerts panel for queries flagged with negative sentiment.